Training Robust Neural Networks

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1 Data preparation

Before training we clean the dataset. Early dropout of weak features is a data-cleaning step: inputs that carry little signal are removed so the model never sees them. This keeps the input space small and the training fast.

2 Regularization

Large networks memorize their training data. Here dropout randomly disables units during training, which forces the network to learn redundant features instead of relying on any single unit. We use a rate of 0.5 in hidden layers and turn it off at evaluation time.

3 Results

With both steps the validation error drops by a clear margin while training time stays roughly the same as the baseline model.